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Homework 03: TCP Congestion Control, IP Addressing, Forwarding, NAT
Points: *100*

- MAY ONLY BE TURNED IN ON **GRADESCOPE as a PDF file.**
- There is NO MAKEUP for missed assignments.
- We are strict about enforcing the LATE POLICY for all assignments (see syllabus).
  Only use the space provided for answers. Use clear and clean handwriting (or typing).
  NOT USING THIS TEMPLATE WILL LOSE YOU 20 POINTS!

1. **(6 pts) TCP and Loss Events.** Complete the following table about loss events in Tahoe vs. Reno. Assume current congestion window is cwnd.

Event	Protocol	New ssthresh	New cwnd	New Phase
3 dup. ACKs	Tahoe	cwnd/2	1 MSS	Slow start
3 dup. ACKs	Reno	cwnd/2	cwnd/2	Congestion avoidance
Timeout	Tahoe	cwnd/2	1 MSS	slow start
Timeout	Reno	cwnd/2	1 MSS	slow start

2. **(14 pts) TCP Congestion Control.** Consider a TCP connection that uses **TCP Reno**. The Maximum Segment Size (MSS) is 1 KB. The current ssthresh is 16 KB. The current cwnd is 12 KB. The connection is currently in the **Congestion Avoidance** phase.

   a. **(3 pts)** If the next 12 KB of data are sent and all are acknowledged successfully, what will the cwnd be at the end of that RTT?

   13 KB

b. (3 pts) Suppose that immediately after the event in Part A, the sender receives three duplicate ACKs.

i. (1 pt) What action does TCP Reno take regarding the ssthresh and cwnd values?

ssthresh + cwnd will be halved

so ssthresh = 6.5 kB
cwnd = 6.5 kB

ii. (1 pt) What phase does the protocol enter next?

Congestion avoidance

iii. (2 pts) How would this differ if the sender was using TCP Tahoe instead?

TCP Tahoe would re-enter slow start and the new cwnd would be 1 MSS = 1 kB instead of 6.5 kB. The new ssthresh is the same however.

c. (4 pts) Later in the same connection, a Timeout occurs when the cwnd is at 24 KB. Describe the state of the cwnd and ssthresh immediately after this timeout, and name the phase the sender enters.

ssthresh = 12 kB
cwnd = 1 kB = 1 MSS

The new phase is slow start since a timeout occured.

d. **(3 pts)** Explain why TCP reacts differently to a Timeout compared to receiving three duplicate ACKs. (Hint: Think about what each event implies about the state of the network "pipe").

3 duplicate ACKs means the receiver is still receiving packets, but some packets got lost / dropped in the network. A timeout means the packet / ACK never reached its intended target indicating high levels of congestion within the network, prompting the sender to re-enter slow start.

3. **(10 pts) Subnets.** A network administrator is given the address block **192.168.10.0/24** and must divide it into subnets for different departments.

a. **(2 pts)** Write the subnet mask for /24 in dotted-decimal notation.

/24 → first 24 bits are all 1's

11111111 . 11111111 . 11111111 . 00000000

= 255.255.255.0

b. **(2 pts)** How many bits are available for host addresses in a /24 network?

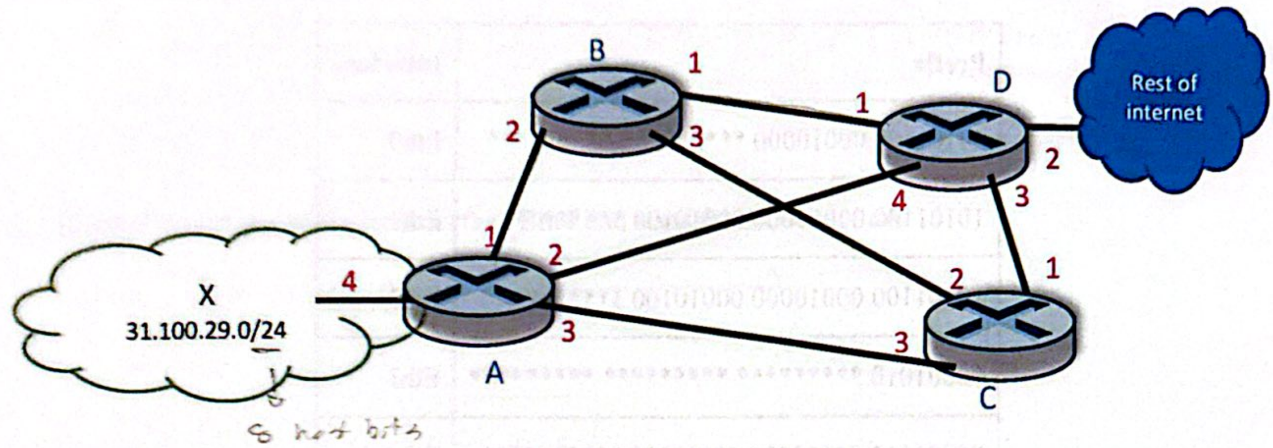
IPv4 has 32 bits, 24 network bits.

32 - 24 = $\boxed{8 \text{ bits}}$

c. **(6 pts) The administrator needs the following subnets. For each, determine the prefix length, subnet mask, network address, broadcast address, and usable host range, given the starting network 192.168.10.0.**

Dept.	Hosts Needed	Prefix Length	Subnet Mask	Network Address	Broadcast Address	Usable Host Range
CS	100	/25	255.255.255.128	192.168.10.0	192.168.10.127	192.168.10.1 – 192.168.10.126
ECE	50 6 host bits	/26	255.255.255.192	192.168.10.128	192.168.10.191	192.168.10.129 – 192.168.10.190
TMP	25 5 host bits	/27	255.255.255.224	192.168.10.192	192.168.10.223	192.168.10.193 – 192.168.10.222
ME	10 4 host bits	/28	255.255.255.240	192.168.10.224	192.168.10.239	192.168.10.225 – 192.168.10.238

4. **(12 pts) Packet Forwarding.** Consider the following network with forwarding tables shown:



A	
Match	**Port**
31.100.29.x	4
100.x.x.x	3
215.0.x.x	2
*	1

B	
Match	**Port**
201.177.8.x	4
86.5.x.x	2
121.7.x.x	1
*	3

C	
Match	**Port**
201.177.8.x	3
86.x.x.x	1
*	2

D	
Match	**Port**
31.x.x.x	4
134.x.x.x	3
86.5.12.x	1
*	2

a. **(3 pts)** A host in network X sends a packet. That packet traverses routers A → B → D. What is one possible remote host IP address the packet could have been sent to?

127.7.1.1

b. **(3 pts)** A packet that started from network X with a TTL of 64 reaches a TTL of 0 and is dropped by router C. What is one possible destination IP address the packet could have had?

100.1.1.1

c. **(3 pts)** What is one possible IP address of a host in network X?

31.100.29.1

d. **(3 pts)** A packet with source address 134.51.2.89 and destination address 31.100.29.11 arrives at router D. What route will the packet take?

D → A → X

5. **(12 pts) Forwarding Tables.** A high-performance backbone router uses the following forwarding table to manage traffic across six outbound interfaces.

Prefix	Interface
10101100 00010000 ******** ********	Eth0
10101100 00010000 00010100 ********	Eth1
10101100 00010000 00010100 1*******	Eth2
00001010 ******** ******** ********	Eth3
00001010 10000000 ******** ********	Eth4
Otherwise	Eth5

a. **(1 pt)** When this router looks at its forwarding table to move a packet from an input port to **Eth2**, is this a **control plane or data plane** function?

data plane function

b. **(6 pts)** A packet arrives with the destination IP 172.16.20.200.

i. **(2 pts)** Convert this IP address into its 32-bit binary representation.

200
-128
72
-64
8

172
-128
44
-32
12
-8
4

10101100.00010000.00010100.
11001000

ii. **(2 pts)** Identify every interface in the table that matches this destination.

Eth0, Eth1, Eth2, Eth5

iii. **(2 pts)** Which specific interface will the router select, and why?

Eth2 because it has the longest matching prefix with the destination IP.

c. **(1 pt)** Calculate the total number of unique IP addresses that belong to the range defined for **Interface Eth2.**

7 undetermined bits => $\boxed{\text{128 unique addresses}}$

d. **(2 pts)** Convert the prefix for **Interface Eth4** into standard CIDR notation.

00001010 10000000

16 defined bits

=> $\boxed{10.128.0.0/16}$

e. **(2 pts)** Suppose a fiber cut occurs and Interface **Eth1** is shut down, removing its entry from the table. If a packet destined for 172.16.20.50 arrives, which interface will now handle the traffic?

10101100 . 00010000 . 00010100 . 00110010

Eth0

6. **(11 pts) NAT.** A small office has a single public IP address **203.0.113.45** assigned by their ISP. Internally, devices use the private subnet **10.0.0.0/24**. The office NAT router maintains a translation table.

a. **(2 pts)** Briefly explain why the office only needs one public IP address for all internal devices to access the internet. What field(s) does the NAT router manipulate to distinguish between connections from different internal hosts?

The NAT router allows multiple internal devices to share a single public IP by multiplexing traffic at the Transport Layer. It manipulates the IP address and source port number to identify connections from different internal hosts.

b. **(6 pts)** Host **10.0.0.5** sends a packet to **142.250.80.46:443**. The router maps it to external port **60001**.

Fill in the src/dst fields at each point in the journey:

- Leaving the internal host
    - **Source IP:** 10.0.0.5
    - **Source Port:** 50200
    - **Destination IP:** 142.250.80.46
    - **Destination Port:** 443
- Outgoing (leaving the router toward the internet)
    - **Source IP:** 203.0.113.45
    - **Source Port:** 60001
    - **Destination IP:** 142.250.80.46
    - **Destination Port:** 443
- Incoming reply (arriving at the router from the internet)
    - **Source IP:** 142.250.80.46
    - **Source Port:** 443
    - **Destination IP:** 203.0.113.45
    - **Destination Port:** 60001
- After NAT translation (delivered to the internal host)
    - **Source IP:** 142.250.80.46
    - **Source Port:** 443
    - **Destination IP:** 10.0.0.5
    - **Destination Port:** 50200

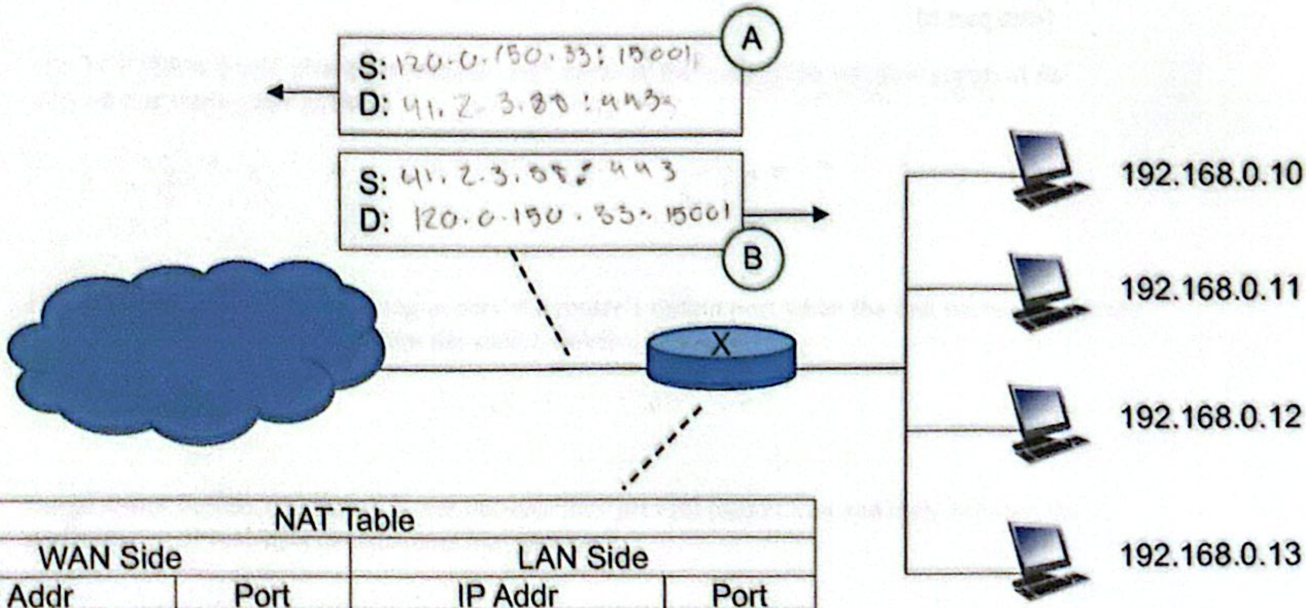
c. **(2 pts)** An external server at 9.9.9.9 wants to initiate a connection to the office's internal web server at 10.0.0.20:8080. Can it do so directly? Why or why not?

No because when the server packet reaches the router, there is no corresponding internal IP/port for it to go to because no request was sent to the server to begin with. The packet is simply dropped.

d. **(1 pt)** Name one technique that can be used to work around this limitation.

Port Forwarding

7. **(12 pts) NAT.** In the diagram below, four hosts are behind a NAT. The public (WAN) address of the router is **120.0.150.33** and the NAT starts at port **15000**. The NAT table starts as shown.



NAT Table			
WAN Side		LAN Side	
IP Addr	Port	IP Addr	Port
120.0.150.33	15000	192.168.0.13	51123
120.0.150.33	15001	192.168.0.10	5555

a. **(2 pts)** The router receives a packet on its LAN interface from 192.168.0.13 destined for port 15000. Does the router need to add a NAT table entry? If so, add it to the table.

No, entry already exists

b. **(6 pts)** A process running on node **192.168.0.10** using port **5555** sends a message to port **443** at host **41.2.3.88.**

   i. Add an entry to the NAT table that results from this message being sent.

   ii. Fill in the headers including the source address, source port, destination address, and destination port:

   - For packet A that the router sends to the remote node.
   - For packet B that the remote host responds with.

c. **(2 pts)** A process running on node **192.168.0.11** using port **5555** sends a message to port **6000** at node **112.3.75.11**. Will the router need to add an entry to the NAT table? (Assume NAT table from part b)

Yes because the internal IP address is different despite using the same port.

d. **(2 pts)** A process running on node **192.168.0.10** using port **5555** sends a message to port **443** at node **112.3.75.11**. Will the router need to add an entry to the NAT table? (Assume NAT table from part b)

No because a mapping from that internal IP/port already exists to an external IP/port.

8. **(8 pts) True or False. Indicate whether each statement is True (T) or False (F).**

a. Flow control is primarily concerned with preventing the sender from overwhelming the network infrastructure (routers and links).

→ congestion control

False

b. DHCP uses TCP to ensure reliable delivery of IP address assignments.

False used UDP

c. A NAT router modifies both the source IP address and port number in outgoing packets.

True

d. TCP Explicit congestion notification (ECN) uses NACKs to notify the sender that a packet was lost.

uses ACKs

False

e. The TCP "Slow Start" phase is named as such because the congestion window grows at its slowest rate during this period.

False

f. Head-of-the-Line (HOL) blocking occurs at a router's output port when the link transmission rate is slower than the arrival rate from the switch fabric.

False happens in input port

g. Large router buffers are always better because they prevent packet loss and thus improve the performance of real-time applications like gaming.

False

h. Both flow control and congestion control result in the sender throttling (slowing down) its transmission rate.

True

(15 pts) Wireshark Lab Handin
Please follow the Wireshark lab writeup and provide your answers to the following questions.

1. **(1 pt)** What is the IP address and TCP port number used by the client computer (source) that is transferring the alice.txt file to gaia.cs.umass.edu?

IP: 192.168.86.68

Port: 55639

2. **(1 pt)** What is the IP address of gaia.cs.umass.edu? On what port number is it sending and receiving TCP segments for this connection?

IP: 128.119.245.12

Port: 80

3. **(2 pts)** What is the sequence number of the TCP SYN segment that is used to initiate the TCP connection between the client computer and gaia.cs.umass.edu? What is it in the segment that identifies the segment as a SYN segment?

4236649187

It has a (rel) sequence number of 0 + doesn't contain an ACK field.

4. **(2 pts)** What is the sequence number of the SYNACK segment sent by gaia.cs.umass.edu to the client computer in reply to the SYN? What is the value of the ACKnowledgement field in the SYNACK segment? How did gaia.cs.umass.edu determine that value?

sequence number: 1068969752

ACK: 1

It determined that value because it is waiting for the 1st byte to be sent by the sender.

5. **(1 pt)** What is the raw (not relative!) sequence number of the TCP segment containing the HTTP POST command?

4236801228

6. **(6 pts)** Consider the TCP segment containing the HTTP POST as the first segment in the TCP connection.
   a. **(1 pt)** At what time was the first segment (the one containing the HTTP POST) in the data transfer part of the TCP connection sent?

   18:43:26.716922

   24.047 ms since first frame

   b. **(1 pt)** At what time was the ACK for this first data-containing segment received?

   18:43:26.745546

   52.671 ms since first frame

   c. **(1 pt)** What is the RTT for this first data-containing segment?

   28.624 ms

   d. **(1 pt)** What is the RTT value of the second data-carrying TCP segment and its ACK?

   28.628 ms

e. **(2 pts)** What is the EstimatedRTT value (see Section 3.5.3, in the text) after the ACK for the second data-carrying segment is received? Assume that in making this calculation after receiving the ACK for the second segment, that the initial value of EstimatedRTT is equal to the measured RTT for the first segment, and then is computed using this EstimatedRTT equation:

$$EstimatedRTT = (1 - \alpha) . EstimatedRTT + \alpha . SampleRTT$$

with a value of $\alpha = 0.125$.

$(1 - 0.125) \cdot 28.624 + 0.125 \cdot 28.628$

$= \boxed{28.625 \text{ ms}}$

7. **(2 pts)** Comment on whether this looks as if TCP is in its slow start phase, congestion avoidance phase or some other phase.

slow start because after the first few RTTs the number of packets sent essentially doubles.